
FULL MOLECULAR HAMILTONIAN

From Classical to Quantum

Chih-En Shen

Contents

1	Classical Cartesian Hamiltonian	4
2	Transformation to the Nuclear Center-of-Mass	4
2.1	New coordinates	4
2.2	Canonical momenta	4
2.3	Hamiltonian after the transformation	5
2.4	Eliminating the Cross-Coupling via a Classical Canonical Transformation . .	5
2.4.1	Generator	6
2.4.2	Induced canonical transformation	6
2.4.3	Canonicity check	7
2.4.4	Transformed Hamiltonian	7
2.4.5	Geometric interpretation	8
3	BF frame for both nuclei and electrons	9
3.1	Setup	9
3.2	From the internal Hamiltonian to a working Routhian	9
3.3	Body-fixed Routhian	10
3.4	Expanding (3.7)	10
3.4.1	Generic squared-term identity	10
3.4.2	Application to the nuclear term	11
3.4.3	Assembled Routhian	11
3.5	Canonical momenta and the total angular momentum	12
3.5.1	Nuclear angular momentum $\mathbf{J}_N = \partial\mathcal{R}/\partial\boldsymbol{\omega}$	12
3.5.2	Total angular momentum $\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}}$	12
3.5.3	Vibrational momentum $P_b = \partial\mathcal{R}/\partial\dot{Q}_b$	12
3.6	Key identity	13
3.6.1	The clean split	13
3.6.2	Cross-check: the $\mathbf{I}_e/\mathbf{I}_X$ cancellation of the velocity-form route	13
3.7	Hamiltonian in canonical variables	14
3.7.1	Nuclear rotation–vibration block	14
3.7.2	Electronic block	15
3.7.3	Final assembly: (3.22)	15
3.8	Complete coupling enumeration	16
4	Quantization	17
4.1	The quantization problem in curvilinear coordinates	17
4.1.1	Podolsky form / Laplace–Beltrami quantization	17
4.2	Volume element and Podolsky rescaling	17
4.2.1	Configuration-space metric and the volume element	17
4.2.2	Wavefunction rescaling	18
4.3	Anomalous BF angular-momentum algebra	19
4.4	Watson pseudopotential	20
4.4.1	Algebraic cross-check of the coefficient	21
4.4.2	Operator ordering: what is and isn’t ambiguous	21

4.5	Electronic kinetic operator	22
4.6	Complete quantum molecular Hamiltonian	22
4.7	Born–Huang expansion	23
4.7.1	Clamped-nuclei electronic Hamiltonian	23
4.7.2	The exact Born–Huang expansion	23
4.7.3	Coupled-channel equations	23
4.7.4	Born–Oppenheimer approximation as truncation	24
4.7.5	Summary	24
4.7.6	Nuclear Center-of-Mass Coordinate	25
4.7.7	Kinetic Energy in Velocity Form	25
4.7.8	Canonical Momenta	26
4.7.9	Kinetic Energy in Momentum Form	27
4.7.10	Expanded Form	28
4.7.11	Hamiltonian	28
4.7.12	Molecular Rest Frame	29
4.7.13	Important Interpretation	29

1 Classical Cartesian Hamiltonian

We have N_{nuc} nuclei labeled by j with masses M_j , charges $Z_j e$, positions $\mathbf{R}_j$, and conjugate momenta $\mathbf{P}_j$; and N_{elec} electrons labeled by i with mass m_e , charge $-e$, positions $\mathbf{r}_i$, and momenta $\mathbf{p}_i$. All vectors live in a space-fixed (SF) lab frame. Using Gaussian units and writing $R_{ij} = |\mathbf{R}_i - \mathbf{R}_j|$ etc.,

$$H = \underbrace{\sum_{j=1}^{N_{\text{nuc}}} \frac{\mathbf{P}_j^2}{2M_j}}_{T_N} + \underbrace{\sum_{i=1}^{N_{\text{elec}}} \frac{\mathbf{p}_i^2}{2m_e}}_{T_e} + V, \quad (1.1)$$

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{i \neq j} \frac{Z_i Z_j e^2}{R_{ij}} - \sum_{i,j} \frac{Z_j e^2}{|\mathbf{R}_j - \mathbf{r}_i|} + \frac{1}{2} \sum_{i \neq j} \frac{e^2}{r_{ij}} \right). \quad (1.2)$$

Canonical Poisson brackets $\{R_{j\alpha}, P_{k\beta}\} = \delta_{jk} \delta_{\alpha\beta}$, $\{r_{i\alpha}, p_{j\beta}\} = \delta_{ij} \delta_{\alpha\beta}$.

2 Transformation to the Nuclear Center-of-Mass

2.1 New coordinates

Define the nuclear COM, $3(N-1)$ internal nuclear vectors, and electron positions in the nuclear COM frame:

$$\mathbf{R}_{\text{cm}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j, \quad M_N = \sum_j M_j, \quad (2.1)$$

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \quad \mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \quad (2.2)$$

The $\mathbf{R}'_j$ are linearly dependent:

$$\sum_j M_j \mathbf{R}'_j = 0. \quad (2.3)$$

2.2 Canonical momenta

This is a point transformation $\mathbf{Q}'_{\text{new}} = \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$ with $\mathbf{q}_{\text{old}} = (\mathbf{R}_j, \mathbf{r}_i)$ and $\mathbf{Q}'_{\text{new}} = (\mathbf{R}_{\text{cm}}, \mathbf{R}'_j, \mathbf{r}'_i)$. Use the generating function: $F_2 = F_2(\mathbf{q}_{\text{old}}, \mathbf{P}, t) = \mathbf{P} \cdot \mathbf{Q}'_{\text{new}}(\mathbf{q}_{\text{old}})$

$$F_2 = \mathbf{P}_{\text{cm}} \cdot \mathbf{R}_{\text{cm}}(\{\mathbf{R}_j\}) + \sum_j \mathbf{P}'_j \cdot \mathbf{R}'_j(\{\mathbf{R}_j\}) + \sum_i \mathbf{p}'_i \cdot \mathbf{r}'_i(\mathbf{R}_j, \mathbf{r}_i). \quad (2.4)$$

Then $\mathbf{P}_j = \partial F_2 / \partial \mathbf{R}_j$, $\mathbf{p}_i = \partial F_2 / \partial \mathbf{r}_i$. Using

$$\frac{\partial \mathbf{R}_{\text{cm}}}{\partial \mathbf{R}_j} = \frac{M_j}{M_N} \mathbb{1}, \quad \frac{\partial \mathbf{R}'_k}{\partial \mathbf{R}_j} = (\delta_{jk} - \frac{M_j}{M_N}) \mathbb{1}, \quad \frac{\partial \mathbf{r}'_i}{\partial \mathbf{R}_j} = -\frac{M_j}{M_N} \mathbb{1}, \quad \frac{\partial \mathbf{r}'_i}{\partial \mathbf{r}_j} = \delta_{ij} \mathbb{1}, \quad (2.5)$$

one obtains

$$\mathbf{p}_i = \mathbf{p}'_i, \quad \mathbf{P}_j = \frac{M_j}{M_N} (\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i - \sum_k \mathbf{P}'_k) + \mathbf{P}'_j. \quad (2.6)$$

The constraint dual to (2.3) is $\sum_k \mathbf{P}'_k = 0$, so

$$\boxed{\mathbf{P}_j = \frac{M_j}{M_N}(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) + \mathbf{P}'_j, \quad \mathbf{p}_i = \mathbf{p}'_i} \quad (2.7)$$

Total momentum: $\sum_j \mathbf{P}_j + \sum_i \mathbf{p}_i = \mathbf{P}_{\text{cm}}$. So $\mathbf{P}_{\text{cm}}$ is the *total* (nuclei+electrons) momentum.

2.3 Hamiltonian after the transformation

Insert (2.7) into T_N in (1.1):

$$T_N = \sum_j \frac{\mathbf{P}_j^2}{2M_j} = \sum_j \frac{1}{2M_j} \left[\frac{M_j}{M_N}(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) + \mathbf{P}'_j \right]^2. \quad (2.8)$$

Expanding the square and summing,

$$T_N = \frac{(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i)^2}{2M_N} + \frac{1}{M_N}(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i) \cdot \underbrace{\sum_j \mathbf{P}'_j}_{=0} + \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j}. \quad (2.9)$$

Combining with T_e and V :

$$H = \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \underbrace{\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}_i}{M_N}}_{\text{artifact}} + \underbrace{\frac{(\sum_i \mathbf{p}_i)^2}{2M_N}}_{\text{mass-polarization}} + \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + V(\mathbf{R}'_j, \mathbf{r}'_i). \quad (2.10)$$

Because H is independent of $\mathbf{R}_{\text{cm}}$, $\mathbf{P}_{\text{cm}}$ is conserved; pick the eigenvalue $\mathbf{P}_{\text{cm}} = 0$ (center-of-mass at rest) and drop the trivial $\mathbf{P}_{\text{cm}}$ -pieces:

$$\boxed{H_{\text{int}} = \sum_j \frac{\mathbf{P}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}_i^2}{2m_e} + \underbrace{\frac{1}{2M_N} \left(\sum_i \mathbf{p}_i \right)^2}_{\text{electron mass polarization}} + V(\mathbf{R}'_j, \mathbf{r}'_i).} \quad (2.11)$$

The potential is unchanged in form because $\mathbf{R}'_j - \mathbf{R}'_i = \mathbf{R}_j - \mathbf{R}_i$, $\mathbf{R}'_j - \mathbf{r}'_i = \mathbf{R}_j - \mathbf{r}_i$, $\mathbf{r}'_i - \mathbf{r}'_j = \mathbf{r}_i - \mathbf{r}_j$. The mass-polarization term, $(M_N)^{-1} \sum_{i < j} \mathbf{p}_i \cdot \mathbf{p}_j + (2M_N)^{-1} \sum_i \mathbf{p}_i^2$, is the price of referring electrons to the nuclear (rather than the total) COM. This rest-frame choice is not essential: Section 2.4 shows that the *same* internal Hamiltonian follows for any $\mathbf{P}_{\text{cm}}$ from a classical canonical transformation, and it is that frame-independent form we carry forward.

2.4 Eliminating the Cross-Coupling via a Classical Canonical Transformation

The rest-frame choice $\mathbf{P}_{\text{cm}} = 0$ made in Section 2.3 is convenient but stronger than necessary, and we now lift it. For $\mathbf{P}_{\text{cm}} \neq 0$ the Hamiltonian (2.10) carries two artifacts inherited from

having referred electrons to the *nuclear* (rather than the *total*) center of mass:

$$H = \underbrace{\frac{\mathbf{P}_{\text{cm}}^2}{2M_N}}_{(A) \text{ wrong total mass}} - \underbrace{\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}'_i}{M_N}}_{(B) \text{ cross-coupling}} + \frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} + \sum_j \frac{\mathbf{p}'_j{}^2}{2M_j} + \sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} + V. \quad (2.12)$$

The wrong-mass term (A) has denominator M_N instead of the proper $M_{\text{tot}} = M_N + N_{\text{elec}}m_e$, and (B) couples the COM motion to the internal electronic momenta. We eliminate both *exactly, for any* $\mathbf{P}_{\text{cm}}$, by a single classical canonical transformation generated on phase space — so that the internal Hamiltonian carried into the body-fixed treatment of Section 3 is the frame-independent object, not a rest-frame special case. This transformation is the classical preimage of the Galilean-boost unitary that one would apply in the quantum theory.

2.4.1 Generator

Define on phase space the function

$$G \equiv \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{r}'_i, \quad M_{\text{tot}} = M_N + N_{\text{elec}}m_e. \quad (2.13)$$

G depends only on $\mathbf{P}_{\text{cm}}$ and $\{\mathbf{r}'_i\}$; in particular it is independent of $\mathbf{R}_{\text{cm}}$, $\mathbf{R}'_j$, $\mathbf{P}'_j$, and $\mathbf{p}'_i$.

2.4.2 Induced canonical transformation

The infinitesimal canonical transformation generated by G acts on any phase-space function f by $\delta f = \{f, G\}$. Because G is linear in $\mathbf{P}_{\text{cm}}$ and $\{\mathbf{r}'_i\}$, the bracket $\{\cdot, G\}$ applied twice annihilates the canonical variables, so the finite transformation at unit parameter is given by a single application:

$$\tilde{f} = f + \{f, G\}. \quad (2.14)$$

Using $\{f, g\} = \sum (\partial_q f \partial_p g - \partial_p f \partial_q g)$, compute the action on each canonical pair.

Electron pair $(\mathbf{r}'_i, \mathbf{p}'_i)$:

$$\begin{aligned} \{r'_{i,a}, G\} &= \frac{\partial G}{\partial p'_{i,a}} = 0, \\ \{p'_{i,a}, G\} &= -\frac{\partial G}{\partial r'_{i,a}} = -\frac{m_e}{M_{\text{tot}}} P_{\text{cm},a}. \end{aligned}$$

Nuclear-COM pair $(\mathbf{R}_{\text{cm}}, \mathbf{P}_{\text{cm}})$:

$$\begin{aligned} \{R_{\text{cm},a}, G\} &= \frac{\partial G}{\partial P_{\text{cm},a}} = \frac{m_e}{M_{\text{tot}}} \sum_i r'_{i,a}, \\ \{P_{\text{cm},a}, G\} &= -\frac{\partial G}{\partial R_{\text{cm},a}} = 0. \end{aligned}$$

Internal nuclear pair $(\mathbf{R}'_j, \mathbf{P}'_j)$: G depends on neither, so $\{\mathbf{R}'_j, G\} = \{\mathbf{P}'_j, G\} = 0$. Hence the finite canonical transformation reads

$$\boxed{\begin{aligned} \tilde{\mathbf{r}}'_i &= \mathbf{r}'_i, & \tilde{\mathbf{p}}'_i &= \mathbf{p}'_i - \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}}, \\ \tilde{\mathbf{R}}_{\text{cm}} &= \mathbf{R}_{\text{cm}} + \frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}'_i, & \tilde{\mathbf{P}}_{\text{cm}} &= \mathbf{P}_{\text{cm}}, \\ \tilde{\mathbf{R}}'_j &= \mathbf{R}'_j, & \tilde{\mathbf{P}}'_j &= \mathbf{P}'_j. \end{aligned}} \quad (2.15)$$

2.4.3 Canonicity check

The only non-obvious Poisson bracket is between the two simultaneously shifted variables:

$$\begin{aligned}
\{\tilde{p}'_{i,a}, \tilde{R}_{\text{cm},b}\} &= \{\mathbf{p}'_{i,a} - \frac{m_e}{M_{\text{tot}}} P_{\text{cm},a}, R_{\text{cm},b} + \frac{m_e}{M_{\text{tot}}} \sum_l r'_{l,b}\} \\
&= \underbrace{\{p'_{i,a}, R_{\text{cm},b}\}}_0 + \frac{m_e}{M_{\text{tot}}} \sum_l \underbrace{\{p'_{i,a}, r'_{l,b}\}}_{-\delta_{il}\delta_{ab}} \\
&\quad - \frac{m_e}{M_{\text{tot}}} \underbrace{\{P_{\text{cm},a}, R_{\text{cm},b}\}}_{-\delta_{ab}} - \frac{m_e^2}{M_{\text{tot}}^2} \sum_l \underbrace{\{P_{\text{cm},a}, r'_{l,b}\}}_0 \\
&= -\frac{m_e}{M_{\text{tot}}} \delta_{ab} + \frac{m_e}{M_{\text{tot}}} \delta_{ab} = 0.
\end{aligned} \tag{2.16}$$

All other elementary brackets are preserved by inspection (each is either unchanged or trivially zero), so (2.15) is canonical.

2.4.4 Transformed Hamiltonian

Invert (2.15) for substitution into (2.12): $\mathbf{p}'_i = \tilde{\mathbf{p}}'_i + (m_e/M_{\text{tot}})\mathbf{P}_{\text{cm}}$. The potential V depends only on $\mathbf{R}'_j$ and $\mathbf{r}'_i$, which are unchanged, so V is form-invariant. Working term by term.

Cross-coupling (B):

$$\begin{aligned}
-\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \mathbf{p}'_i}{M_N} &= -\frac{\mathbf{P}_{\text{cm}} \cdot (\sum_i \tilde{\mathbf{p}}'_i + N_{\text{elec}} \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}})}{M_N} \\
&= -\frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N} - \frac{N_{\text{elec}} m_e \mathbf{P}_{\text{cm}}^2}{M_N M_{\text{tot}}}.
\end{aligned} \tag{2.17}$$

Mass-polarization quadratic:

$$\begin{aligned}
\frac{(\sum_i \mathbf{p}'_i)^2}{2M_N} &= \frac{(\sum_i \tilde{\mathbf{p}}'_i + N_{\text{elec}} \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}})^2}{2M_N} \\
&= \frac{(\sum_i \tilde{\mathbf{p}}'_i)^2}{2M_N} + \frac{N_{\text{elec}} m_e \mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}^2 m_e^2 \mathbf{P}_{\text{cm}}^2}{2M_N M_{\text{tot}}^2}.
\end{aligned} \tag{2.18}$$

Electron kinetic energy:

$$\begin{aligned}
\sum_i \frac{\mathbf{p}'_i{}^2}{2m_e} &= \sum_i \frac{(\tilde{\mathbf{p}}'_i + \frac{m_e}{M_{\text{tot}}} \mathbf{P}_{\text{cm}})^2}{2m_e} \\
&= \sum_i \frac{\tilde{\mathbf{p}}'_i{}^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i}{M_{\text{tot}}} + \frac{N_{\text{elec}} m_e \mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}^2}.
\end{aligned} \tag{2.19}$$

Coefficient of $\mathbf{P}_{\text{cm}} \cdot \sum_i \tilde{\mathbf{p}}'_i$: Summing (2.17), (2.18), (2.19),

$$-\frac{1}{M_N} + \frac{N_{\text{elec}} m_e}{M_N M_{\text{tot}}} + \frac{1}{M_{\text{tot}}} = \frac{-M_{\text{tot}} + N_{\text{elec}} m_e + M_N}{M_N M_{\text{tot}}} = 0, \tag{2.20}$$

using $M_{\text{tot}} = M_N + N_{\text{elec}} m_e$. *The cross-coupling (B) is killed identically.*

Coefficient of $\mathbf{P}_{\text{cm}}^2$: Adding the $\mathbf{P}_{\text{cm}}^2/(2M_N)$ from (A) plus the residual $\mathbf{P}_{\text{cm}}^2$ -pieces from

(2.17)–(2.19),

$$\begin{aligned}
\frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}^2 m_e^2}{2M_N M_{\text{tot}}^2} + \frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} &= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} \left(\frac{N_{\text{elec}}m_e}{M_N} + 1 \right) \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}m_e}{2M_{\text{tot}}^2} \cdot \frac{M_{\text{tot}}}{M_N} \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{M_N M_{\text{tot}}} + \frac{N_{\text{elec}}m_e}{2M_N M_{\text{tot}}} \\
&= \frac{1}{2M_N} - \frac{N_{\text{elec}}m_e}{2M_N M_{\text{tot}}} = \frac{M_{\text{tot}} - N_{\text{elec}}m_e}{2M_N M_{\text{tot}}} = \frac{M_N}{2M_N M_{\text{tot}}} = \frac{1}{2M_{\text{tot}}}.
\end{aligned} \tag{2.21}$$

The wrong-mass (A) is corrected to M_{tot} .

Assembling everything,

$$\boxed{\tilde{H} = \frac{\mathbf{P}_{\text{cm}}^2}{2M_{\text{tot}}} + \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\tilde{\mathbf{p}}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \tilde{\mathbf{p}}_i' \right)^2 + V(\mathbf{R}_j', \mathbf{r}_i').} \tag{2.22}$$

The COM kinetic energy carries the *correct* total mass M_{tot} . The internal piece is structurally identical to (2.11): the same mass-polarization $(\sum_i \tilde{\mathbf{p}}_i')^2/(2M_N)$, the same nuclear kinetic energy, the same potential. Since $\tilde{H}$ is independent of $\tilde{\mathbf{R}}_{\text{cm}}$, $\tilde{\mathbf{P}}_{\text{cm}} = \mathbf{P}_{\text{cm}}$ is conserved, and one may proceed with the internal dynamics for *any* value of $\mathbf{P}_{\text{cm}}$ — not just zero — with no kinetic artifact.

2.4.5 Geometric interpretation

The new “COM coordinate” is

$$\tilde{\mathbf{R}}_{\text{cm}} = \mathbf{R}_{\text{cm}} + \frac{m_e}{M_{\text{tot}}} \sum_i \mathbf{r}_i' = \frac{M_N \mathbf{R}_{\text{cm}} + m_e \sum_i \mathbf{r}_i}{M_{\text{tot}}} = \mathbf{R}_{\text{tot}}^{\text{CoM}}, \tag{2.23}$$

using $\mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}_i'$. The canonical transformation has *quietly relabeled* the COM coordinate from nuclear to total COM, while *leaving the relative coordinates $\mathbf{R}_j'$ and $\mathbf{r}_i'$ untouched*. This is precisely what makes the framework still “nuclear-COM-based”: the Born–Oppenheimer-friendly relative coordinates of Section 2.1 survive unchanged. The momentum shift $\tilde{\mathbf{p}}_i' = \mathbf{p}_i' - (m_e/M_{\text{tot}})\mathbf{P}_{\text{cm}}$ is the kinetic compensation required to keep the Poisson brackets canonical: in Galilean terms, each electron’s lab-frame momentum is reduced by $m_e \mathbf{V}_{\text{CoM}}$ with $\mathbf{V}_{\text{CoM}} = \mathbf{P}_{\text{cm}}/M_{\text{tot}}$.

The canonical transformation has thus cleanly separated the total COM motion *exactly and for any $\mathbf{P}_{\text{cm}}$* : in (2.22) the COM appears only through the decoupled free term $\mathbf{P}_{\text{cm}}^2/2M_{\text{tot}}$ — carrying the *correct* total mass M_{tot} and free of the cross-coupling (B) — while every other term depends only on the internal variables. It is this total Hamiltonian (2.22), exact for *any* $\mathbf{P}_{\text{cm}}$ and not merely the rest frame, that we carry into the body-fixed treatment of Section 3.

3 BF frame for both nuclei and electrons

3.1 Setup

With Eckart's frame fixed by the *nuclei*,

$$\mathbf{R}'_j = \mathbf{S}(\phi, \theta, \chi) \bar{\mathbf{R}}_j(Q), \quad \mathbf{r}'_i = \mathbf{S}(\phi, \theta, \chi) \bar{\mathbf{r}}_i. \quad (3.1)$$

The Eckart conditions $\sum_j M_j \bar{\mathbf{R}}_j = 0$ and $\sum_j M_j \bar{\mathbf{R}}_j^{\text{eq}} \times \bar{\mathbf{R}}_j = 0$ fix the orientation of $\mathbf{S}$ from the *nuclear* positions; the same $\mathbf{S}$ is then used to define BF electron coordinates $\bar{\mathbf{r}}_i$. The nuclear shape is parametrized by rectilinear mass-weighted **normal coordinates** Q_b ,

$$\bar{\mathbf{R}}_j(Q) = \bar{\mathbf{R}}_j^{\text{eq}} + \frac{1}{\sqrt{M_j}} \sum_b \mathbf{l}_{jb} Q_b, \quad (3.1a)$$

with mass-weighted normal-mode vectors $\mathbf{l}_{jb}$ obeying orthonormality and the Eckart (Sayvetz) conditions

$$\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}, \quad \sum_j \sqrt{M_j} \mathbf{l}_{jb} = 0, \quad \sum_j \sqrt{M_j} \bar{\mathbf{R}}_j^{\text{eq}} \times \mathbf{l}_{jb} = 0, \quad (3.1b)$$

so that $\partial \bar{\mathbf{R}}_j / \partial Q_b = \mathbf{l}_{jb} / \sqrt{M_j}$ is constant.

BF angular velocity:

$$(\mathbf{S}^T \dot{\mathbf{S}})_{\alpha\beta} = -\epsilon_{\alpha\beta\gamma} \omega_\gamma \implies \mathbf{S}^T \dot{\mathbf{S}} \mathbf{v} = \boldsymbol{\omega} \times \mathbf{v}. \quad (3.2)$$

Velocities (using $|\mathbf{S}\mathbf{v}|^2 = |\mathbf{v}|^2$):

$$\dot{\mathbf{R}}'_j = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j), \quad \dot{\mathbf{r}}'_i = \mathbf{S}(\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i). \quad (3.3)$$

3.2 From the internal Hamiltonian to a working Routhian

I need an action principle for the body-fixed *nuclear* coordinates. The electrons, however, are best left in the canonical (momentum) form they already carry in (2.22): their kinetic energy and mass polarization form a rotational scalar that passes through the body-fixed transformation unchanged. I therefore Legendre-transform *only* the COM and nuclear momenta back to velocities — keeping $\mathbf{P}_{\text{cm}}$ arbitrary, never setting it to zero — and keep the electron momenta $\tilde{\mathbf{p}}'_i$ as canonical variables. The relevant Hamilton equations from (2.22) are

$$\dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} / M_{\text{tot}}, \quad \dot{\mathbf{R}}'_j = \mathbf{P}'_j / M_j, \quad \dot{\mathbf{r}}'_i = \tilde{\mathbf{p}}'_i / m_e + (\sum_l \tilde{\mathbf{p}}'_l) / M_N, \quad (3.4)$$

the last of which inverts to the electron momentum–velocity relation [recorded for the velocity-form cross-check below, around (3.18)]

$$\tilde{\mathbf{p}}'_i = m_e \dot{\mathbf{r}}'_i - \frac{m_e^2}{M_{\text{tot}}} \sum_l \dot{\mathbf{r}}'_l, \quad \sum_l \tilde{\mathbf{p}}'_l = \frac{m_e M_N}{M_{\text{tot}}} \sum_i \dot{\mathbf{r}}'_i. \quad (3.5)$$

Legendre-transforming the COM and nuclear momenta only, $\mathcal{R} = \mathbf{P}_{\text{cm}} \dot{\tilde{\mathbf{R}}}_{\text{cm}} + \sum_j \mathbf{P}'_j \dot{\tilde{\mathbf{R}}}'_j - \tilde{H}$, yields the **Routhian**

$$\boxed{\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\tilde{\mathbf{R}}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\dot{\tilde{\mathbf{R}}}'_j|^2 - H_e(\tilde{\mathbf{p}}'_i) - V, \quad H_e \equiv \sum_i \frac{\tilde{\mathbf{p}}'^2_i}{2m_e} + \frac{1}{2M_N} \left(\sum_i \tilde{\mathbf{p}}'_i \right)^2,} \quad (3.6)$$

which is a Lagrangian in $(\tilde{\mathbf{R}}_{\text{cm}}, \mathbf{R}'_j)$ and *minus* a Hamiltonian in $(\mathbf{r}'_i, \tilde{\mathbf{p}}'_i)$: extremizing $\int \mathcal{R} dt$ gives Euler–Lagrange equations for the COM and nuclei and Hamilton’s equations for the electrons. The electronic kinetic energy and mass polarization thus retain their (2.22) momentum form H_e — they are never converted to velocity form. (The overall minus sign on H_e is the Routhian’s partial-Legendre signature; the mass polarization keeps its physical positive coefficient $+1/2M_N$ inside H_e , and the whole H_e re-enters the Hamiltonian with a plus sign when it is reassembled in (3.22).)

3.3 Body-fixed Routhian

Substitute the body-fixed velocities (3.3) into the nuclear term of (3.6); the COM term is untouched by the rotation. The electron Hamiltonian H_e is a *rotational scalar* — it is built from the dot products $\tilde{\mathbf{p}}'_i \cdot \tilde{\mathbf{p}}'_i$ — so under the passive rotation to the body frame it is unchanged in form. Defining the body-fixed electron momenta $\bar{\mathbf{p}}_i \equiv \mathbf{S}^T \tilde{\mathbf{p}}'_i$ (a proper rotation, so $|\bar{\mathbf{p}}_i|^2 = |\tilde{\mathbf{p}}'_i|^2$ and $\sum_i \bar{\mathbf{p}}_i = \mathbf{S}^T \sum_i \tilde{\mathbf{p}}'_i$, whence $(\sum_i \bar{\mathbf{p}}_i)^2 = (\sum_i \tilde{\mathbf{p}}'_i)^2$),

$$H_e = \sum_i \frac{\tilde{\mathbf{p}}'^2_i}{2m_e} + \frac{1}{2M_N} \left(\sum_i \tilde{\mathbf{p}}'_i \right)^2 = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2. \quad (3.6a)$$

Defining $\bar{\mathbf{r}} \equiv \sum_i \bar{\mathbf{r}}_i$, the body-fixed Routhian is therefore

$$\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\tilde{\mathbf{R}}}_{\text{cm}}|^2 + \frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\tilde{\mathbf{R}}}'_j|^2 - \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} - \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 - V, \quad (3.7)$$

in which only the nuclear kinetic term carries the body-fixed angular velocity $\boldsymbol{\omega}$. The potential is now manifestly **S**-independent:

$$V = \frac{1}{4\pi\epsilon_0} \left(\frac{1}{2} \sum_{j \neq k} \frac{Z_j Z_k e^2}{R_{jk}} - \sum_{j,i} \frac{Z_j e^2}{|\bar{\mathbf{R}}_j - \bar{\mathbf{r}}_i|} + \frac{1}{2} \sum_{i \neq l} \frac{e^2}{\bar{r}_{il}} \right). \quad (3.8)$$

Rotational symmetry is rendered explicit; the BF electronic problem at fixed Q is self-contained.

3.4 Expanding (3.7)

3.4.1 Generic squared-term identity

For any vectors $\mathbf{a}, \dot{\mathbf{a}}, \boldsymbol{\omega} \in \mathbb{R}^3$, the BCH-style identity $\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega} a^2 - \mathbf{a}(\mathbf{a} \cdot \boldsymbol{\omega})$ gives

$$\begin{aligned} |\boldsymbol{\omega} \times \mathbf{a}|^2 &= (\boldsymbol{\omega} \times \mathbf{a}) \cdot (\boldsymbol{\omega} \times \mathbf{a}) = \boldsymbol{\omega} \cdot [\mathbf{a} \times (\boldsymbol{\omega} \times \mathbf{a})] \\ &= \boldsymbol{\omega} \cdot [\boldsymbol{\omega} a^2 - \mathbf{a}(\mathbf{a} \cdot \boldsymbol{\omega})] = \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}, \end{aligned} \quad (3.9a)$$

while the scalar triple product gives

$$(\boldsymbol{\omega} \times \mathbf{a}) \cdot \dot{\mathbf{a}} = \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}}). \quad (3.9b)$$

Hence, for any constant mass-like coefficient m ,

$$\begin{aligned} \frac{m}{2} |\boldsymbol{\omega} \times \mathbf{a} + \dot{\mathbf{a}}|^2 &= \frac{m}{2} [|\boldsymbol{\omega} \times \mathbf{a}|^2 + 2(\boldsymbol{\omega} \times \mathbf{a}) \cdot \dot{\mathbf{a}} + |\dot{\mathbf{a}}|^2] \\ &= \underbrace{\frac{m}{2} \boldsymbol{\omega}^T (a^2 \mathbb{1} - \mathbf{a} \mathbf{a}^T) \boldsymbol{\omega}}_{\text{rotational}} + \underbrace{m \boldsymbol{\omega} \cdot (\mathbf{a} \times \dot{\mathbf{a}})}_{\text{Coriolis-type}} + \underbrace{\frac{m}{2} |\dot{\mathbf{a}}|^2}_{\text{vibrational}}. \end{aligned} \quad (3.9c)$$

3.4.2 Application to the nuclear term

Only the nuclear kinetic term of the Routhian (3.7) carries $\boldsymbol{\omega}$. Apply (3.9c) with $m \rightarrow M_j$, $\mathbf{a} \rightarrow \bar{\mathbf{R}}_j$ and sum over j :

$$\frac{1}{2} \sum_j M_j |\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\bar{\mathbf{R}}}_j|^2 = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N. \quad (3.9d)$$

The electronic kinetic energy is *not* expanded this way — it is carried as the rotational scalar H_e of (3.6a). Only for the velocity-form cross-check of § 3.6 do we record the decomposition that the electronic and mass-polarization terms *would* produce if the electrons were instead kept in velocity form,

$$\frac{m_e}{2} \sum_i |\boldsymbol{\omega} \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i|^2 = \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_e \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_e + T_e^{\text{BF}}, \quad -\frac{m_e^2}{2M_{\text{tot}}} |\boldsymbol{\omega} \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}}|^2 = -\frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_X \boldsymbol{\omega} - \boldsymbol{\omega} \cdot \mathbf{L}_X - T_X^{\text{mp}}. \quad (3.9e)$$

The constituent inertia tensors are

$$\mathbf{I}_N = \sum_j M_j (\bar{R}_j^2 \mathbb{1} - \bar{\mathbf{R}}_j \bar{\mathbf{R}}_j^T), \quad \mathbf{I}_e = m_e \sum_i (\bar{r}_i^2 \mathbb{1} - \bar{\mathbf{r}}_i \bar{\mathbf{r}}_i^T), \quad (3.9)$$

$$\mathbf{I}_X = \frac{m_e^2}{M_{\text{tot}}} (\bar{r}^2 \mathbb{1} - \bar{\mathbf{r}} \bar{\mathbf{r}}^T), \quad (3.10)$$

the angular-momentum-like (velocity-form) quantities are

$$\mathbf{L}_N^{\text{vib}} = \sum_j M_j \bar{\mathbf{R}}_j \times \dot{\bar{\mathbf{R}}}_j, \quad \mathbf{L}_e = m_e \sum_i \bar{\mathbf{r}}_i \times \dot{\bar{\mathbf{r}}}_i, \quad \mathbf{L}_X = \frac{m_e^2}{M_{\text{tot}}} \bar{\mathbf{r}} \times \dot{\bar{\mathbf{r}}}, \quad (3.11)$$

and the “vibrational” kinetic pieces are

$$T_{\text{vib}}^N = \frac{1}{2} \sum_j M_j |\dot{\bar{\mathbf{R}}}_j|^2 = \frac{1}{2} \sum_b \dot{Q}_b^2, \quad T_e^{\text{BF}} = \frac{m_e}{2} \sum_i |\dot{\bar{\mathbf{r}}}_i|^2, \quad T_X^{\text{mp}} = \frac{m_e^2}{2M_{\text{tot}}} |\dot{\bar{\mathbf{r}}}|^2. \quad (3.12)$$

3.4.3 Assembled Routhian

Inserting the nuclear expansion (3.9d) into (3.7), together with the decoupled COM term, the electron scalar H_e , and the potential:

$$\boxed{\mathcal{R} = \frac{1}{2} M_{\text{tot}} |\dot{\bar{\mathbf{R}}}_{\text{cm}}|^2 + \underbrace{\frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N}_{\text{nuclear, velocity form}} - H_e - V.} \quad (3.13)$$

The rotational kernel is the **nuclear** inertia $\mathbf{I}_N$ *alone*: because the electrons are carried as the scalar H_e rather than expanded in $\boldsymbol{\omega}$, no electronic or mass-polarization inertia ($\mathbf{I}_e, \mathbf{I}_X$) enters the kernel. The electrons couple to rotation only *kinematically*, through their angular momentum, which we identify next. The body-fixed electron momenta and the **electronic angular momentum** are

$$\bar{\mathbf{p}}_i = \mathbf{S}^T \tilde{\mathbf{p}}'_i, \quad \mathbf{L}_{\text{elec}} \equiv \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i. \quad (3.14)$$

Remark. The electronic and mass-polarization inertia/angular-momentum quantities $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ of (3.9)–(3.11), and the velocity-form decomposition (3.9e), are needed *only* in the cross-check of § 3.6; they play no role in the Routhian (3.13) itself.

3.5 Canonical momenta and the total angular momentum

Compute the conjugate momenta from the Routhian $\mathcal{R}$. The cyclic COM coordinate gives the conserved total momentum $\mathbf{P}_{\text{cm}} = \partial\mathcal{R}/\partial\dot{\mathbf{R}}_{\text{cm}} = M_{\text{tot}}\dot{\mathbf{R}}_{\text{cm}}$; being decoupled, it contributes nothing to the internal momenta below. The electron momenta $\bar{\mathbf{p}}_i$ are already canonical [eq. (3.14)]; it remains to find the rotational and vibrational momenta, most transparently by differentiating the unexpanded nuclear term of (3.7).

3.5.1 Nuclear angular momentum $\mathbf{J}_N = \partial\mathcal{R}/\partial\boldsymbol{\omega}$

In the Routhian (3.13) only the nuclear rotational kernel and the nuclear Coriolis piece carry $\boldsymbol{\omega}$ — the electron scalar H_e and the COM term do not. The momentum conjugate to the body-fixed angular velocity is therefore the *nuclear* angular momentum

$$\mathbf{J}_N \equiv \frac{\partial\mathcal{R}}{\partial\boldsymbol{\omega}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}}. \quad (3.15)$$

3.5.2 Total angular momentum $\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}}$

The body-fixed frame $\mathbf{S}(\Omega)$ rotates the nuclei *and* the electrons together [eq. (3.1)], so the total angular momentum conjugate to the Euler angles — the physical, conserved rovibronic angular momentum — is the sum of the nuclear and electronic contributions. The electronic part is the canonical electron angular momentum $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ of (3.14), the body-fixed image of $\sum_i \mathbf{r}'_i \times \tilde{\mathbf{p}}'_i$ (the cross product co-rotates with $\mathbf{S}$). Hence

$$\mathbf{J} = \mathbf{J}_N + \mathbf{L}_{\text{elec}} = \mathbf{I}_N \boldsymbol{\omega} + \mathbf{L}_N^{\text{vib}} + \mathbf{L}_{\text{elec}}. \quad (3.16)$$

3.5.3 Vibrational momentum $P_b = \partial\mathcal{R}/\partial\dot{Q}_b$

Only $\dot{\mathbf{R}}_j$ carries $\dot{Q}_b$ dependence; with $\partial\bar{\mathbf{R}}_j/\partial Q_b = \mathbf{l}_{jb}/\sqrt{M_j}$ from (3.1a),

$$P_b \equiv \frac{\partial\mathcal{R}}{\partial\dot{Q}_b} = \sum_j M_j (\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\mathbf{R}}_j) \cdot \frac{\partial\bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j} (\boldsymbol{\omega} \times \bar{\mathbf{R}}_j + \dot{\mathbf{R}}_j) \cdot \mathbf{l}_{jb}. \quad (3.17)$$

Separating the rotational and vibrational pieces defines the **Coriolis vectors** σ_b ,

$$\sigma_b \equiv \sum_j M_j \bar{\mathbf{R}}_j \times \frac{\partial \bar{\mathbf{R}}_j}{\partial Q_b} = \sum_j \sqrt{M_j} \bar{\mathbf{R}}_j \times \mathbf{l}_{jb} = \sum_a \zeta_{ab} Q_a, \quad \zeta_{ab} \equiv \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}, \quad (3.17a)$$

the second form following from (3.1a) with the Eckart condition $\sum_j \sqrt{M_j} \bar{\mathbf{R}}_j^{\text{eq}} \times \mathbf{l}_{jb} = 0$. The vibrational metric is the identity, $\sum_j M_j (\partial \bar{\mathbf{R}}_j / \partial Q_a) \cdot (\partial \bar{\mathbf{R}}_j / \partial Q_b) = \sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}$, so the canonical vibrational momentum carries *no* Wilson G -matrix,

$$P_b = \omega \cdot \sigma_b(Q) + \dot{Q}_b. \quad (3.17b)$$

3.6 Key identity

3.6.1 The clean split

The total-angular-momentum decomposition (3.16) *is* the key identity: subtracting the electronic part from both sides,

$$\boxed{\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N(Q) \omega + \mathbf{L}_N^{\text{vib}}}. \quad (3.19)$$

The right side is a purely nuclear-velocity quantity, with the *nuclear* inertia $\mathbf{I}_N$ as rotational kernel. The entire electronic and mass-polarization content sits on the left, in the kinematic shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}}$; it never enters the kernel, because in the Routhian (3.13) the electrons were carried as the scalar H_e and contributed no inertia at all.

3.6.2 Cross-check: the $\mathbf{I}_e/\mathbf{I}_X$ cancellation of the velocity-form route

Had we instead converted the electrons to velocity form — expanding their kinetic energy via (3.9e) — the electronic inertia and Coriolis terms would have entered *both* $\mathbf{J}$ and $\mathbf{L}_{\text{elec}}$, and (3.19) would emerge only after an exact cancellation. The velocity-form electron momentum is $\bar{\mathbf{p}}_i = m_e(\omega \times \bar{\mathbf{r}}_i + \dot{\bar{\mathbf{r}}}_i) - \frac{m_e^2}{M_{\text{tot}}}(\omega \times \bar{\mathbf{r}} + \dot{\bar{\mathbf{r}}})$ [the body-fixed image of (3.5)]; substituting into $\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ and using the BAC–CAB identity $\mathbf{a} \times (\omega \times \mathbf{a}) = (a^2 \mathbf{1} - \mathbf{a} \mathbf{a}^T) \omega$ on the two ω -terms collapses the four pieces to

$$\mathbf{L}_{\text{elec}} = (\mathbf{I}_e - \mathbf{I}_X) \omega + (\mathbf{L}_e - \mathbf{L}_X), \quad (3.18)$$

with $\mathbf{I}_e, \mathbf{I}_X, \mathbf{L}_e, \mathbf{L}_X$ from (3.9)–(3.11). The velocity-form total angular momentum would read $\mathbf{J} = (\mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X) \omega + (\mathbf{L}_N^{\text{vib}} + \mathbf{L}_e - \mathbf{L}_X)$ [from (3.9d)+(3.9e)], so

$$\mathbf{J} - \mathbf{L}_{\text{elec}} = \mathbf{I}_N \omega + \mathbf{L}_N^{\text{vib}} + \underbrace{[(\mathbf{I}_e - \mathbf{I}_X) \omega + (\mathbf{L}_e - \mathbf{L}_X)] - \mathbf{L}_{\text{elec}}}_{= 0 \text{ by (3.18)}} = \mathbf{I}_N \omega + \mathbf{L}_N^{\text{vib}},$$

reproducing (3.19): the electronic inertia $\mathbf{I}_e$ and the mass-polarization inertia $\mathbf{I}_X$ cancel *exactly*. This “algebraic miracle” of the Eckart construction is automatic in the mixed treatment, where $\mathbf{I}_e$ and $\mathbf{I}_X$ never arise.

3.6.3 Inversion: the Watson reciprocal inertia

To invert (3.19) for $\boldsymbol{\omega}$ we must re-express the velocity-form $\mathbf{L}_N^{\text{vib}}$ in canonical variables. In normal coordinates $\mathbf{L}_N^{\text{vib}} = \sum_j M_j \dot{\mathbf{R}}_j \times \dot{\mathbf{R}}_j = \sum_b \boldsymbol{\sigma}_b \dot{Q}_b$; eliminating $\dot{Q}_b = P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b$ from (3.17b),

$$\mathbf{L}_N^{\text{vib}} = \sum_b \boldsymbol{\sigma}_b (P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b) = \boldsymbol{\pi} - \boldsymbol{\sigma} \boldsymbol{\sigma}^T \boldsymbol{\omega}, \quad \boldsymbol{\pi} \equiv \sum_b \boldsymbol{\sigma}_b P_b, \quad \boldsymbol{\sigma} \boldsymbol{\sigma}^T \equiv \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T, \quad (3.20a)$$

with $\boldsymbol{\pi}$ the **field-free vibrational angular momentum**. Substituting into (3.19), $\mathbf{J} - \mathbf{L}_{\text{elec}} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T) \boldsymbol{\omega} + \boldsymbol{\pi} \equiv \mathbf{I}' \boldsymbol{\omega} + \boldsymbol{\pi}$, so

$$\boldsymbol{\omega} = \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}), \quad \boxed{\boldsymbol{\mu} \equiv (\mathbf{I}')^{-1} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}}, \quad (3.20)$$

the **Watson reciprocal-inertia tensor**. The rotation–vibration coupling is now carried by the modified inertia $\mathbf{I}'$ and the uniform shift $\mathbf{J} \rightarrow \mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}$ — with no Wilson G -matrix.

3.7 Hamiltonian in canonical variables

We now assemble the body-fixed Hamiltonian in the canonical variables $(\Omega, \mathbf{J}; Q, P_b; \bar{\mathbf{r}}_i, \bar{\mathbf{p}}_i)$. The total energy splits as $\tilde{H} = \mathbf{P}_{\text{cm}}^2 / 2M_{\text{tot}} + H_{\text{int}}$, the COM piece being a decoupled spectator. The internal Hamiltonian is the sum of the nuclear rotation–vibration kinetic energy, the electron scalar, and the potential,

$$H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V, \quad T_N^{\text{rot+vib}} \equiv \frac{1}{2} \boldsymbol{\omega}^T \mathbf{I}_N \boldsymbol{\omega} + \boldsymbol{\omega} \cdot \mathbf{L}_N^{\text{vib}} + T_{\text{vib}}^N,$$

read off the Routhian (3.13). The electron scalar H_e [eq. (3.6a)] and the potential V are already canonical; only $T_N^{\text{rot+vib}}$ must be re-expressed through the momenta $(\mathbf{J}, P_b)$.

3.7.1 Nuclear rotation–vibration block

$T_N^{\text{rot+vib}}$ is homogeneous-quadratic in the nuclear velocities $(\boldsymbol{\omega}, \dot{Q}_b)$, so by Euler’s theorem $2T_N^{\text{rot+vib}} = \mathbf{J}_N \cdot \boldsymbol{\omega} + \sum_b P_b \dot{Q}_b$, with conjugate momenta $\mathbf{J}_N = \mathbf{J} - \mathbf{L}_{\text{elec}}$ [eqs. (3.15), (3.19)] and P_b [eq. (3.17b)]. Substituting $\dot{Q}_b = P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b$ and $\sum_b P_b \boldsymbol{\sigma}_b = \boldsymbol{\pi}$,

$$2T_N^{\text{rot+vib}} = (\mathbf{J} - \mathbf{L}_{\text{elec}}) \cdot \boldsymbol{\omega} + \sum_b P_b (P_b - \boldsymbol{\omega} \cdot \boldsymbol{\sigma}_b) = (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) \cdot \boldsymbol{\omega} + \sum_b P_b^2.$$

Inserting $\boldsymbol{\omega} = \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})$ from (3.20),

$$\boxed{T_N^{\text{rot+vib}} \Big|_{\text{canonical}} = \frac{1}{2} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) + \frac{1}{2} \sum_b P_b^2.} \quad (3.20d)$$

The vibrational kinetic energy is the *bare* $\frac{1}{2} \sum_b P_b^2$; the entire rotation–vibration coupling is carried by $\boldsymbol{\pi}$ and by the Watson inertia $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}$ — no Wilson G -matrix survives.

3.7.2 Electronic block

The electronic kinetic energy and mass polarization require no further work: they are the rotational scalar H_e of (3.6a), carried unchanged from the Step-2 Hamiltonian (2.22),

$$H_e = \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2. \quad (3.21)$$

Because H_e is built from the rotation-invariant dot products $\bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_i$ and $\bar{\mathbf{p}}_i = \mathbf{S}^T \tilde{\mathbf{p}}'_i$ is a passive rotation, its body-fixed form is *identical* to the Step-2 form (2.22) under $\tilde{\mathbf{p}}'_i \rightarrow \bar{\mathbf{p}}_i$ — no Legendre transform or angular-momentum expansion is needed. The mass polarization survives the body-fixed transformation intact in functional form.

Reduced-mass check ($N_{\text{elec}} = 1$): $H_e = \bar{\mathbf{p}}_1^2/(2m_e) + \bar{\mathbf{p}}_1^2/(2M_N) = \bar{\mathbf{p}}_1^2/(2\mu_e)$ with $\mu_e = m_e M_N / (m_e + M_N) = m_e M_N / M_{\text{tot}}$. ✓

3.7.3 Final assembly: (3.22)

Add (3.20d) and (3.21), include $V = V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$ from (3.8), and use $H_{\text{int}} = T_N^{\text{rot+vib}} + H_e + V$:

$$\begin{aligned} H_{\text{int}} = & \frac{1}{2} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu}(Q) (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) \\ & + \frac{1}{2} \sum_b P_b^2 \\ & + \sum_i \frac{\bar{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \bar{\mathbf{p}}_i \right)^2 \\ & + V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\}). \end{aligned} \quad (3.22)$$

Restoring the decoupled COM piece, the complete canonical Hamiltonian is $\tilde{H} = \mathbf{P}_{\text{cm}}^2 / 2M_{\text{tot}} + H_{\text{int}}$ with H_{int} given by (3.22) — exact for *any* $\mathbf{P}_{\text{cm}}$. The COM term is a conserved free-particle spectator that decouples from every body-fixed variable, so the quantization of Section 4 proceeds with H_{int} .

Here $\boldsymbol{\pi} = \sum_b \boldsymbol{\sigma}_b P_b$ is the field-free vibrational angular momentum, with the Coriolis vectors $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ of (3.17a) and constants $\boldsymbol{\zeta}_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$; the vibrational metric being the identity, no Wilson G -matrix enters.

$\mathbf{L}_{\text{elec}} = \sum_i \bar{\mathbf{r}}_i \times \bar{\mathbf{p}}_i$ is the BF canonical electronic angular momentum, and crucially:

- $\boldsymbol{\mu} = (\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)^{-1}$ is the **Watson reciprocal-inertia tensor**, built from nuclear quantities alone: *no* electron or mass-polarization corrections enter the rotational kernel — they were absorbed into $\mathbf{L}_{\text{elec}}$ via (3.19).
- The electron contribution to rotation appears entirely as the Coriolis-like coupling $-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$.
- The mass polarization is preserved in its $\left(\sum_i \bar{\mathbf{p}}_i \right)^2 / (2M_N)$ form.

3.8 Complete coupling enumeration

Expanding the rotational kernel:

$$\frac{1}{2}(\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi})^T \boldsymbol{\mu} (\mathbf{J} - \mathbf{L}_{\text{elec}} - \boldsymbol{\pi}) =$$

$$\underbrace{\frac{1}{2}\mathbf{J}^T \boldsymbol{\mu} \mathbf{J}}_{\text{rotation}} \underbrace{-\mathbf{J}^T \boldsymbol{\mu} \boldsymbol{\pi}}_{\text{rot-vib}} \underbrace{-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}}_{\text{Coriolis rot-elec}} \underbrace{+\boldsymbol{\pi}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}}_{\text{Coriolis vib-elec}} + \frac{1}{2}\boldsymbol{\pi}^T \boldsymbol{\mu} \boldsymbol{\pi} + \frac{1}{2}\mathbf{L}_{\text{elec}}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}.$$

All couplings in (3.22):

1. **Pure rotation, anisotropic top:** $\frac{1}{2}\mathbf{J}^T \boldsymbol{\mu}(Q)\mathbf{J}$ — Q -dependent moments of inertia (rot-vib coupling through inertia).
2. **Rot-vib Coriolis:** $-\mathbf{J}^T \boldsymbol{\mu} \boldsymbol{\pi}$.
3. **Rot-electron Coriolis:** $-\mathbf{J}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$ — the kinetic source of Λ -doubling, Renner-Teller, spin-rotation analogues.
4. **Vib-electron angular coupling:** $\boldsymbol{\pi}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$ — exact non-adiabatic angular coupling intrinsic to BF embedding.
5. **Quadratic vibrational angular momentum:** $\frac{1}{2}\boldsymbol{\pi}^T \boldsymbol{\mu} \boldsymbol{\pi}$.
6. **Quadratic electronic angular momentum:** $\frac{1}{2}\mathbf{L}_{\text{elec}}^T \boldsymbol{\mu} \mathbf{L}_{\text{elec}}$.
7. **Pure vibrational kinetic:** $\frac{1}{2}\sum_b P_b^2$.
8. **Electronic kinetic:** $\sum_i \bar{\mathbf{p}}_i^2/(2m_e)$.
9. **Mass-polarization, diagonal:** $\sum_i \bar{\mathbf{p}}_i^2/(2M_N)$ — modifies effective electron mass.
10. **Mass-polarization, off-diagonal (two-electron):** $\sum_{i<l} \bar{\mathbf{p}}_i \cdot \bar{\mathbf{p}}_l/M_N$.
11. **Coulomb:** $V_{NN}(Q) + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\})$. All Euler-angle independent.

Compared to the previous (electrons-in-SF) framework: kinetic rot-electron coupling has *replaced* potential rot-electron coupling. V_{Ne} is now manifestly $\mathbf{S}$ -independent — the canonical chemistry advantage.

4 Quantization

The classical internal Hamiltonian (3.22) is parametrized by curvilinear coordinates (Euler angles ϕ, θ, χ and vibrational normal coordinates Q_b) and BF Cartesian electron coordinates $\bar{\mathbf{r}}_i$. Canonical quantization in curvilinear coordinates requires care: the naïve substitution $\pi \rightarrow -i\hbar\partial$ does not produce a Hermitian operator on the natural Hilbert space. The Podolsky (Laplace–Beltrami) prescription resolves this; the residual ordering produces the Watson pseudopotential, and the BF angular momentum acquires an *anomalous* commutator algebra.

4.1 The quantization problem in curvilinear coordinates

Label the configuration-space coordinates q^A :

$$\{q^A\} = \{\phi, \theta, \chi, Q_1, \dots, Q_{3N_{\text{nucl}}-6}, \bar{\mathbf{r}}_1, \dots, \bar{\mathbf{r}}_{N_{\text{elec}}}\}, \quad (4.1)$$

with conjugate momenta π_A . The classical kinetic energy from (3.22) is a quadratic form

$$T = \frac{1}{2} g^{AB}(q) \pi_A \pi_B, \quad (4.2)$$

where g^{AB} is the inverse of the configuration-space metric g_{AB} defined by $T = \frac{1}{2} g_{AB} \dot{q}^A \dot{q}^B$. The metric inherits block structure from (3.22): a rotational block governed by the Watson reciprocal inertia $\boldsymbol{\mu}(Q) = (\mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^T)^{-1}$, an identity vibrational block (normal coordinates), and a Cartesian block for $\bar{\mathbf{r}}_i$.

4.1.1 Podolsky form / Laplace–Beltrami quantization

The natural Hilbert space is $L^2(\mathcal{C}, \sqrt{|g|} d^n q)$, where the Riemannian volume element $\sqrt{|g|} d^n q$ is the unique measure compatible with the metric. To produce a Hermitian kinetic operator on this Hilbert space, replace T by the Laplace–Beltrami operator divided by $-2/\hbar^2$:

$$\boxed{\hat{T} = -\frac{\hbar^2}{2} \frac{1}{\sqrt{|g|}} \partial_A (\sqrt{|g|} g^{AB} \partial_B).} \quad (4.3)$$

This is the *Podolsky prescription*. Equivalently, one may write $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|})$ and then take $\hat{T} = \frac{1}{2} \hat{\pi}_A g^{AB} \hat{\pi}_B$ in symmetric (Weyl-type) ordering; the two prescriptions coincide up to terms absorbed into the volume element.

For our problem, only the Euler-angle and Q_b sectors are curvilinear; the BF Cartesian electronic sector retains the standard $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\bar{\mathbf{r}}_i}$ on $L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}} \bar{\mathbf{r}})$.

4.2 Volume element and Podolsky rescaling

4.2.1 Configuration-space metric and the volume element

The configuration-space metric requires the kinetic energy with *all* coordinates — the electrons included — carried as velocities $\mathbf{v} = (\boldsymbol{\omega}, \dot{\mathbf{Q}}, \dot{\bar{\mathbf{r}}}_i)$. (This velocity-form representation, used here solely to read off the measure, is distinct from the Routhian (3.13), in which the electrons are kept in momentum form.) Expanding the body-fixed nuclear, electronic, and

mass-polarization kinetic terms via (3.9d)–(3.9e), in normal coordinates (vibrational block $\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}$),

$$2T = \mathbf{v}^T \mathbf{M} \mathbf{v}, \quad \mathbf{M} = \begin{pmatrix} \mathbf{I} & \mathbf{Z} & \mathbf{C} \\ \mathbf{Z}^T & \mathbb{1} & 0 \\ \mathbf{C}^T & 0 & \mathbf{D} \end{pmatrix}, \quad (4.3a)$$

with $\mathbf{I} = \mathbf{I}_N + \mathbf{I}_e - \mathbf{I}_X$ the full instantaneous inertia ($\boldsymbol{\omega}$ – $\boldsymbol{\omega}$ block), $\mathbf{Z}_{\alpha b} = (\boldsymbol{\sigma}_b)_\alpha$ the rotation–vibration Coriolis block, $\mathbf{C}$ the rotation–electron block (from $\boldsymbol{\omega} \cdot (\mathbf{L}_e - \mathbf{L}_X)$), and $\mathbf{D}_{(i\alpha)(j\beta)} = (m_e \delta_{ij} - m_e^2/M_{\text{tot}}) \delta_{\alpha\beta}$ the *constant* electron block. The physical velocities relate to the coordinate velocities $\dot{q} = (\dot{\phi}, \dot{\theta}, \dot{\chi}, \dot{Q}_b, \dot{\mathbf{r}}_i)$ by $\mathbf{v} = \mathbf{A}(q)\dot{q}$ with $\mathbf{A} = \text{diag}(\mathbf{E}(\Omega), \mathbb{1}, \mathbb{1})$, where $\boldsymbol{\omega} = \mathbf{E}(\Omega)\dot{\Omega}$ is the Euler kinematic relation. Hence $g_{AB} = (\mathbf{A}^T \mathbf{M} \mathbf{A})_{AB}$ and

$$\det g = (\det \mathbf{A})^2 \det \mathbf{M} = \sin^2 \theta \det \mathbf{M}, \quad |\det \mathbf{E}(\Omega)| = \sin \theta. \quad (4.3b)$$

Reduce $\det \mathbf{M}$ by the Schur complement of the constant electron block $\mathbf{D}$:

$$\det \mathbf{M} = \det \mathbf{D} \det \begin{pmatrix} \mathbf{I} - \mathbf{C} \mathbf{D}^{-1} \mathbf{C}^T & \mathbf{Z} \\ \mathbf{Z}^T & \mathbb{1} \end{pmatrix}. \quad (4.3c)$$

The electron Schur correction is precisely the electronic + mass-polarization inertia $\mathbf{I}_e - \mathbf{I}_X$ of (3.9e): completing the square in $\dot{\mathbf{r}}_i$ (equivalently, eliminating the electron velocities in favour of the momenta $\bar{\mathbf{p}}_i$) removes all $\boldsymbol{\omega}$ -dependence from the electron block, $\mathbf{C} \mathbf{D}^{-1} \mathbf{C}^T = \mathbf{I}_e - \mathbf{I}_X$, so $\mathbf{I} - \mathbf{C} \mathbf{D}^{-1} \mathbf{C}^T = \mathbf{I}_N$ — the same $\mathbf{I}_e/\mathbf{I}_X$ cancellation as the cross-check (3.18). A second Schur complement, now of the $\mathbb{1}$ vibrational block, gives

$$\det \mathbf{M} = \det \mathbf{D} \det(\mathbf{I}_N - \mathbf{Z} \mathbf{Z}^T) = \det \mathbf{D} \det \mathbf{I}', \quad \mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T, \quad (4.3d)$$

using $\mathbf{Z} \mathbf{Z}^T = \sum_b \boldsymbol{\sigma}_b \boldsymbol{\sigma}_b^T = \boldsymbol{\sigma} \boldsymbol{\sigma}^T$, with $\det \mathbf{D}$ a constant. The Coriolis coupling therefore enters the determinant through the *same* modified inertia $\mathbf{I}'$ that defines the kernel $\boldsymbol{\mu} = (\mathbf{I}')^{-1}$. Dropping the constant electron factor (an overall normalization of the flat electronic measure), the Riemannian volume element is

$$\boxed{\sqrt{|g|} d^n q = \underbrace{\sin \theta d\phi d\theta d\chi}_{\text{Haar on } SO(3)} \cdot \underbrace{\sqrt{\det \mathbf{I}'(Q)} \prod_b dQ_b}_{\text{rot. - vib. Jacobian}} \cdot \underbrace{\prod_i d^3 \bar{\mathbf{r}}_i}_{\text{flat (const.)}}.} \quad (4.4)$$

The Jacobian carries the Watson modified inertia $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T$, *not* the bare $\mathbf{I}_N$: the rotation–vibration Coriolis coupling reduces the rotational determinant from $\det \mathbf{I}_N$ to $\det(\mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^T)$.

4.2.2 Wavefunction rescaling

To work with the simpler measure $d\mu_W \equiv \sin \theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3 \bar{\mathbf{r}}_i$, define the rescaled wavefunction

$$\psi_W \equiv (\det \mathbf{I}'(Q))^{1/4} \psi. \quad (4.5)$$

Then $|\psi|^2 \sqrt{|g|} d^n q = |\psi_W|^2 d\mu_W$, so $\psi_W \in L^2(\mathcal{C}, d\mu_W)$. The kinetic operator transforms by a similarity: $\hat{T}_W = \mathcal{J}^{1/2} \hat{T} \mathcal{J}^{-1/2}$ with $\mathcal{J} = (\det \mathbf{I}')^{1/2}$. The residual non-cancellation

generates a new Q -dependent ordinary potential — the Watson pseudopotential (next subsection).

4.3 Anomalous BF angular-momentum algebra

Total angular momentum is naturally defined as an SF observable; its BF components are obtained by projection through $\mathbf{S}$:

$$\hat{J}_\alpha^{\text{BF}} = \mathbf{S}_{\beta\alpha} \hat{J}_\beta^{\text{SF}}. \quad (4.6)$$

The space-fixed components are ordinary angular-momentum operators in the Euler angles; they obey the normal algebra and act on the direction-cosine matrix $\mathbf{S} = \mathbf{S}(\Omega)$ through its *space-fixed* index,

$$[\hat{J}_\gamma^{\text{SF}}, \hat{J}_\delta^{\text{SF}}] = i\hbar \epsilon_{\gamma\delta\lambda} \hat{J}_\lambda^{\text{SF}}, \quad [\hat{J}_\gamma^{\text{SF}}, S_{\delta\beta}] = i\hbar \epsilon_{\gamma\delta\lambda} S_{\lambda\beta}. \quad (4.6a)$$

Because $\mathbf{S}$ is a proper rotation ($\det \mathbf{S} = 1$), the triple product of direction cosines collapses to one Levi-Civita symbol, and orthogonality inverts the projection (4.6):

$$\epsilon_{\gamma\delta\lambda} S_{\gamma\alpha} S_{\delta\beta} S_{\lambda\nu} = \det(\mathbf{S}) \epsilon_{\alpha\beta\nu} = \epsilon_{\alpha\beta\nu}, \quad \hat{J}_\lambda^{\text{SF}} = S_{\lambda\nu} \hat{J}_\nu^{\text{BF}}. \quad (4.6b)$$

Now evaluate the body-fixed commutator (writing $\hat{J} \equiv \hat{J}^{\text{SF}}$), moving the Euler-angle functions $S_{\delta\beta}$ leftward past $\hat{J}_\gamma$ with (4.6a):

$$\begin{aligned} [\hat{J}_\alpha^{\text{BF}}, \hat{J}_\beta^{\text{BF}}] &= [S_{\gamma\alpha} \hat{J}_\gamma, S_{\delta\beta} \hat{J}_\delta] = S_{\gamma\alpha} S_{\delta\beta} [\hat{J}_\gamma, \hat{J}_\delta] + S_{\gamma\alpha} [\hat{J}_\gamma, S_{\delta\beta}] \hat{J}_\delta - S_{\delta\beta} [\hat{J}_\delta, S_{\gamma\alpha}] \hat{J}_\gamma \\ &= i\hbar \epsilon_{\gamma\delta\lambda} (S_{\gamma\alpha} S_{\delta\beta} \hat{J}_\lambda + S_{\gamma\alpha} S_{\lambda\beta} \hat{J}_\delta + S_{\delta\beta} S_{\lambda\alpha} \hat{J}_\gamma), \end{aligned}$$

where $[\hat{J}_\gamma, \hat{J}_\delta] = i\hbar \epsilon_{\gamma\delta\lambda} \hat{J}_\lambda$, $[\hat{J}_\gamma, S_{\delta\beta}] = i\hbar \epsilon_{\gamma\delta\lambda} S_{\lambda\beta}$, and $-\hat{J}_\delta, S_{\gamma\alpha}] = -i\hbar \epsilon_{\delta\gamma\lambda} S_{\lambda\alpha} = +i\hbar \epsilon_{\gamma\delta\lambda} S_{\lambda\alpha}$ were collected under a common $\epsilon_{\gamma\delta\lambda}$. Insert $\hat{J} = \mathbf{S} \hat{J}^{\text{BF}}$ in each term and apply (4.6b):

$$\epsilon_{\gamma\delta\lambda} S_{\gamma\alpha} S_{\delta\beta} \hat{J}_\lambda = \epsilon_{\alpha\beta\nu} \hat{J}_\nu^{\text{BF}}, \quad \epsilon_{\gamma\delta\lambda} S_{\gamma\alpha} S_{\lambda\beta} \hat{J}_\delta = \epsilon_{\alpha\nu\beta} \hat{J}_\nu^{\text{BF}}, \quad \epsilon_{\gamma\delta\lambda} S_{\delta\beta} S_{\lambda\alpha} \hat{J}_\gamma = \epsilon_{\nu\beta\alpha} \hat{J}_\nu^{\text{BF}}.$$

Since $\epsilon_{\alpha\beta\nu} + \epsilon_{\alpha\nu\beta} + \epsilon_{\nu\beta\alpha} = \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} - \epsilon_{\alpha\beta\nu} = -\epsilon_{\alpha\beta\nu}$, the three pieces sum to a single term of *reversed* sign — the extra contribution from differentiating $\mathbf{S}$ outweighs and flips the bare SF algebra:

$$\boxed{[\hat{J}_\alpha^{\text{BF}}, \hat{J}_\beta^{\text{BF}}] = -i\hbar \epsilon_{\alpha\beta\gamma} \hat{J}_\gamma^{\text{BF}}.} \quad (4.7)$$

This is the famous **anomalous commutator** for BF angular momentum (Klein, Van Vleck, 1929; reviewed in Bunker–Jensen).

In contrast, BF electronic angular momentum is built intrinsically from BF Cartesian electron operators:

$$[\hat{L}_{\text{elec},\alpha}, \hat{L}_{\text{elec},\beta}] = +i\hbar \epsilon_{\alpha\beta\gamma} \hat{L}_{\text{elec},\gamma}, \quad (4.8)$$

the *normal* algebra. Since $\hat{J}^{\text{BF}}$ acts only on Euler angles and $\hat{\mathbf{L}}_{\text{elec}}$ acts only on BF electron coordinates, the two commute:

$$[\hat{J}_\alpha^{\text{BF}}, \hat{L}_{\text{elec},\beta}] = 0. \quad (4.9)$$

The vibrational $\hat{\pi}_\alpha$, built from Q_b and $\hat{P}_b$, also commutes with both $\hat{J}^{\text{BF}}$ and $\hat{\mathbf{L}}_{\text{elec}}$. From

here on we write $\hat{J}_\alpha \equiv \hat{J}_\alpha^{\text{BF}}$ and the anomalous algebra (4.7) is understood.

4.4 Watson pseudopotential

Apply the Podolsky prescription (4.3) to the Watson Hamiltonian (3.22) sector by sector, followed by the rescaling (4.5).

Rotational sector is self-adjoint as written. The kernel $\frac{1}{2} \sum_{\alpha\beta} \hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta$, $\hat{A}_\alpha \equiv \hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha$, contains $\mu_{\alpha\beta}(Q)$, which commutes with $\hat{J}$ (Euler-angle vs. Q sectors) and with $\hat{\mathbf{L}}_{\text{elec}}$ (BF electron coordinates). Since $\mu_{\alpha\beta} = \mu_{\beta\alpha}$,

$$(\hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta)^\dagger = \hat{A}_\beta \mu_{\alpha\beta} \hat{A}_\alpha = \hat{A}_\alpha \mu_{\beta\alpha} \hat{A}_\beta = \hat{A}_\alpha \mu_{\alpha\beta} \hat{A}_\beta,$$

so the rotational kernel is Hermitian on $\sin\theta d\Omega$ and contributes *no* extrapotential; the anomalous algebra (4.7) merely fixes the ordering of the $\hat{A}_\alpha \hat{A}_\beta$ and is absorbed once the symmetric form is adopted.

Vibrational sector generates the pseudopotential. In normal coordinates the vibrational metric is flat ($g^{ab} = \delta_{ab}$) but the volume element carries $\gamma(Q) \equiv \det \mathbf{I}'(Q)$ from (4.4). The Podolsky vibrational operator is therefore

$$\hat{T}_{\text{vib}} = -\frac{\hbar^2}{2} \gamma^{-1/2} \sum_b \partial_{Q_b} (\gamma^{1/2} \partial_{Q_b}). \quad (4.10a)$$

The rescaling $\psi_W = \gamma^{1/4} \psi$ of (4.5) sends $\hat{T}_{\text{vib}} \rightarrow \gamma^{1/4} \hat{T}_{\text{vib}} \gamma^{-1/4}$. Expanding (the $\frac{1}{4}(\partial_{Q_b} \ln \gamma) \partial_{Q_b}$ cross terms cancel),

$$\begin{aligned} \gamma^{1/4} \hat{T}_{\text{vib}} \gamma^{-1/4} &= -\frac{\hbar^2}{2} \sum_b \left[\partial_{Q_b}^2 - \frac{1}{4} (\partial_{Q_b} \ln \gamma)^2 - \frac{1}{4} \partial_{Q_b}^2 \ln \gamma \right] \\ &= \frac{1}{2} \sum_b \hat{P}_b^2 + U_{\text{vib}}, \quad \hat{P}_b = -i\hbar \partial_{Q_b}, \end{aligned} \quad (4.10b)$$

where $\hat{P}_b$ is now Hermitian on the flat measure $\prod_b dQ_b$ and the extrapotential is the compact ‘‘Pauli’’ form

$$U_{\text{vib}} = \frac{\hbar^2}{8} \sum_b \left[\partial_{Q_b}^2 \ln \gamma + \frac{1}{4} (\partial_{Q_b} \ln \gamma)^2 \right] = \frac{\hbar^2}{2} \gamma^{-1/4} \left(\sum_b \partial_{Q_b}^2 \right) \gamma^{1/4}. \quad (4.10c)$$

Collecting the Hermitian rotational kernel, the rescaled vibrational term, and the residue,

$$\begin{aligned} \hat{T}_{\text{rovib}} &= \frac{1}{2} \sum_{\alpha\beta} (\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha) \mu_{\alpha\beta}(Q) (\hat{J}_\beta - \hat{L}_{\text{elec},\beta} - \hat{\pi}_\beta) \\ &\quad + \frac{1}{2} \sum_b \hat{P}_b^2 + \hat{U}(Q). \end{aligned} \quad (4.10)$$

Watson collapse to the trace term. The residue U_{vib} of (4.10c), together with the cross-sector pieces of the Podolsky operator and the reordering of $\hat{\pi}$ against $\mu(Q)$, simplifies dramatically. Because $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma} \boldsymbol{\sigma}^\top$ is *quadratic* in the coordinates through the linear

Coriolis vectors $\sigma_b = \sum_a \zeta_{ab} Q_a$ of (3.17a), Jacobi's formula $\partial_{Q_b} \ln \gamma = \sum_{\alpha\beta} \mu_{\alpha\beta} \partial_{Q_b} \mathbf{I}'_{\alpha\beta}$ expresses every derivative through the *constant* Coriolis matrices $\zeta_{ab} = \sum_j \mathbf{l}_{ja} \times \mathbf{l}_{jb}$; their contraction sum rules then make all Q -derivative terms cancel against one another, and the total reduces to the celebrated compact **Watson trace** (Watson, 1968; the full reduction is carried out in Appendix B):

$$\hat{U}(Q) = -\frac{\hbar^2}{8} \sum_{\alpha=1}^3 \mu_{\alpha\alpha}(Q), \quad \mu = (\mathbf{I}_N - \sigma\sigma^\top)^{-1}. \quad (4.11)$$

The electrons do not modify $\hat{U}$: $\mu_{\alpha\beta}$ depends on Q only, the BF-electronic sector enters the rovib block solely through $\hat{\mathbf{L}}_{\text{elec}}$ (which commutes with both μ and the Q -measure γ), and the electronic Cartesian Jacobian is flat. The pseudopotential is a c -number on the nuclear configuration space, fixed entirely by the modified inertia $\mathbf{I}'$.

4.4.1 Algebraic cross-check of the coefficient

The $-\hbar^2/8$ is confirmed by the equivalent algebraic (Weyl-ordering) route noted under (4.3): with the Hermitian momenta $\hat{\pi}_A = -i\hbar(\partial_A + \frac{1}{2}\partial_A \ln \sqrt{|g|})$ one has $\hat{T} = \frac{1}{2}\hat{\pi}_A g^{AB} \hat{\pi}_B$, identical to the Laplace–Beltrami operator. Three observations fix the result:

- *Vibrational sector.* With $g^{ab} = \delta_{ab}$ and $\sqrt{|g|} \propto \gamma^{1/2}$, expanding $\frac{1}{2} \sum_b \hat{\pi}_{Q_b}^2$ (the Hermitian momenta of (4.3) specialized to the modes) and rescaling reproduces exactly the residue (4.10c), $U_{\text{vib}} = \frac{\hbar^2}{2} \gamma^{-1/4} (\sum_b \partial_{Q_b}^2) \gamma^{1/4}$ — the $-\hbar^2/8$ scale enters through the two $\frac{1}{2} \partial_A \ln \sqrt{|g|}$ factors, $\frac{1}{2} \hat{\pi}_{Q_b}^2 \supset \frac{\hbar^2}{8} (\partial_{Q_b} \ln \gamma)^2$ etc.
- *Rotational sector.* $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$, so the only reordering is among the $\hat{J}$'s; by the anomalous algebra (4.7) the antisymmetric part of $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$ vanishes ($\mu_{\alpha\beta} = \mu_{\beta\alpha}$), so this block adds no independent c -number.
- *Reduction.* Carrying the Watson cancellations (Jacobi's formula and the Coriolis sum rules) through both sectors collapses every Q -derivative to the single trace $\hat{U} = -\frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}$, in agreement with (4.11).

The pseudopotential is a c -number depending only on Q (a function on the nuclear configuration space).

4.4.2 Operator ordering: what is and isn't ambiguous

- $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$ (different coordinate sectors: Euler angles vs. Q), with $\hat{\mathbf{L}}_{\text{elec}}$ (BF electron coords), but not with $\hat{P}_b$ (since $\partial \mu_{\alpha\beta} / \partial Q_b \neq 0$). Hence $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta = \mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta$.
- Within the rotational block, the symmetric product $\hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$ in (4.10) is unambiguous because μ is symmetric in (α, β) , and the Watson pseudopotential (4.11) accounts for the residual non-commutativity of the $\hat{J}$'s themselves.
- In normal coordinates the vibrational term is the bare $\frac{1}{2} \sum_b \hat{P}_b^2$; the Q -dependence of the volume element has been moved into $\hat{U}(Q)$ by the rescaling (4.5), which is precisely what Podolsky produces.

4.5 Electronic kinetic operator

In BF Cartesian electron coordinates, $\hat{\mathbf{p}}_i = -i\hbar\nabla_{\mathbf{r}_i}$ is Hermitian on $L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}}\mathbf{r})$. The electronic + mass-polarization kinetic operator is therefore the straightforward quantization of (3.21):

$$\hat{T}_e = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2. \quad (4.12)$$

Expanding the second piece:

$$\frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 = \sum_i \frac{\hat{\mathbf{p}}_i^2}{2M_N} + \frac{1}{M_N} \sum_{i<l} \hat{\mathbf{p}}_i \cdot \hat{\mathbf{p}}_l. \quad (4.13)$$

The diagonal $1/(2M_N)$ piece renormalizes the effective electron mass through $1/m_e + 1/M_N = M_{\text{tot}}/(m_e M_N) = 1/\mu_e$. The off-diagonal two-electron term is the mass-polarization *sensu stricto*, dynamically coupling electrons through nuclear recoil — a purely classical effect (Section 2.4) inherited by the quantum theory.

4.6 Complete quantum molecular Hamiltonian

Combining (4.10), (4.11), (4.12), and the rotationally invariant Coulomb potentials from (3.8):

$$\begin{aligned} \hat{H}_{\text{int}} = & \frac{1}{2} \sum_{\alpha\beta} (\hat{J}_\alpha - \hat{L}_{\text{elec},\alpha} - \hat{\pi}_\alpha) \mu_{\alpha\beta}(Q) (\hat{J}_\beta - \hat{L}_{\text{elec},\beta} - \hat{\pi}_\beta) \\ & + \frac{1}{2} \sum_b \hat{P}_b^2 \\ & + \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 \\ & - \frac{\hbar^2}{8} \sum_\alpha \mu_{\alpha\alpha}(Q) \\ & + V_{NN}(Q) + V_{Ne}(Q, \{\mathbf{r}_i\}) + V_{ee}(\{\mathbf{r}_i\}), \end{aligned} \quad (4.14)$$

acting on $L^2(\mathcal{C}, d\mu_W)$ with $d\mu_W = \sin\theta d\phi d\theta d\chi \prod_b dQ_b \prod_i d^3\mathbf{r}_i$. All operators are Hermitian on this Hilbert space.

This is the **exact non-relativistic molecular Hamiltonian** after separation of the total COM motion, in BF + Eckart frame, with electrons referred to the nuclear COM. The only approximations made along the way were: (i) non-relativistic dynamics; (ii) Eckart's choice of the BF orientation (a convention, not an approximation); (iii) parametrization of the nuclear shape by $3N_{\text{nuc}} - 6$ internal coordinates Q_b (a parametrization, also exact).

4.7 Born–Huang expansion

4.7.1 Clamped-nuclei electronic Hamiltonian

At each nuclear configuration Q , define the **BF clamped-nuclei electronic Hamiltonian** as the sum of all Q -parametric electron-only pieces of (4.14):

$$\hat{H}_{\text{el}}(Q) \equiv \sum_i \frac{\hat{\mathbf{p}}_i^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \hat{\mathbf{p}}_i \right)^2 + V_{Ne}(Q, \{\bar{\mathbf{r}}_i\}) + V_{ee}(\{\bar{\mathbf{r}}_i\}). \quad (4.15)$$

Remark. The mass-polarization piece $(\sum_i \hat{\mathbf{p}}_i)^2/(2M_N)$ is included here as a small correction with $1/M_N$. Some treatments (“clamped-nuclei BO”) drop it; we keep it explicit because it is parametric in Q only through the absent dependence (i.e., M_N is just a number), which means it acts trivially on the parametric family of electronic Hilbert spaces.

Solve the parametric eigenvalue problem:

$$\hat{H}_{\text{el}}(Q) |\phi_n(Q)\rangle = E_n^{\text{el}}(Q) |\phi_n(Q)\rangle, \quad (4.16)$$

in the electronic Hilbert space $\mathcal{H}_e = L^2(\mathbb{R}^{3N_{\text{elec}}}, d^{3N_{\text{elec}}} \bar{\mathbf{r}})$. The eigenstates $\{|\phi_n(Q)\rangle\}$ depend *parametrically* on Q — Q is treated as a fixed classical parameter, not a quantum degree of freedom — and form a complete orthonormal basis of $\mathcal{H}_e$ at each Q :

$$\sum_n |\phi_n(Q)\rangle \langle \phi_n(Q)| = \hat{1}_e, \quad \langle \phi_m(Q) | \phi_n(Q) \rangle_e = \delta_{mn}. \quad (4.17)$$

4.7.2 The exact Born–Huang expansion

The *exact* molecular wavefunction $|\Psi\rangle \in L^2(\mathcal{C}, d\mu_W)$ admits a resolution against the parametric electronic basis. Insert the resolution of identity (4.17) at each Q :

$$|\Psi\rangle = \sum_n |\chi_n\rangle \otimes |\phi_n(Q)\rangle, \quad \chi_n(\Omega, Q) \equiv \langle \phi_n(Q) | \Psi \rangle_e, \quad (4.18)$$

where $\Omega = (\phi, \theta, \chi)$, the electronic inner product is taken at fixed Q , and χ_n are the **nuclear coefficient functions**. The expansion (4.18) is **exact**, not an ansatz: it follows from the completeness (4.17) at each Q , with the only subtlety being that the basis $|\phi_n(Q)\rangle$ varies parametrically. There is *no* simple tensor factorization $\mathcal{H}_{\text{molecule}} = \mathcal{H}_N \otimes \mathcal{H}_e$ because of this parametric dependence; the Born–Huang expansion is the rigorous replacement.

4.7.3 Coupled-channel equations

Substitute (4.18) into $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$, project from the left by $\langle \phi_m(Q)|$, and use (4.16):

$$[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)] \chi_n(\Omega, Q) + \sum_m \hat{\Lambda}_{nm} \chi_m(\Omega, Q) = E \chi_n(\Omega, Q), \quad (4.19)$$

where $\hat{T}_N$ denotes the nuclear (rotational + vibrational) kinetic operator from the first two lines of (4.14). The **non-adiabatic coupling operators** $\hat{\Lambda}_{nm}$ arise from the action of $\hat{T}_N$ (and the rot–electronic couplings through $\hat{\mathbf{L}}_{\text{elec}}$) on the Q -parametric dependence of

$|\phi_m(Q)\rangle$:

$$\hat{\Lambda}_{nm} \sim -\frac{\hbar^2}{1} \langle \phi_n | \nabla_Q \phi_m \rangle_e \cdot \nabla_Q - \frac{\hbar^2}{2} \langle \phi_n | \nabla_Q^2 \phi_m \rangle_e + \dots, \quad (4.20)$$

with analogous terms from the rotation–electron Coriolis $\hat{J}_\alpha \hat{L}_{\text{elec},\alpha}$ kernel.

4.7.4 Born–Oppenheimer approximation as truncation

The **Born–Oppenheimer (BO) approximation** is the diagonal truncation of (4.19): set $\hat{\Lambda}_{nm} \rightarrow 0$ for $m \neq n$ and (optionally) also $\hat{\Lambda}_{nn} \rightarrow 0$. The resulting effective nuclear problem is

$$[\hat{T}_N + E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q)] \chi_n^{\text{BO}}(\Omega, Q) = E^{\text{BO}} \chi_n^{\text{BO}}(\Omega, Q), \quad (4.21)$$

where the **Born–Oppenheimer potential energy surface** for electronic state n is

$$V_n^{\text{BO}}(Q) \equiv E_n^{\text{el}}(Q) + V_{NN}(Q) + \hat{U}(Q). \quad (4.22)$$

The exactness of (4.18) and the well-definedness of (4.19) are mathematical facts. BO is the physical assumption that off-diagonal $\hat{\Lambda}_{nm}$ are small — justified when electronic gaps are large compared to nuclear kinetic energies, breaking down at conical intersections and avoided crossings, where the parametric variation of $|\phi_n(Q)\rangle$ becomes singular.

4.7.5 Summary

1. (4.14) is the exact non-relativistic quantum molecular Hamiltonian in BF + Eckart + nuclear-COM coordinates.
2. The Born–Huang expansion (4.18) is exact, following from completeness of the parametric electronic basis (4.17).
3. The infinite coupled system (4.19) is equivalent to the Schrödinger equation $\hat{H}_{\text{int}}|\Psi\rangle = E|\Psi\rangle$.
4. The Born–Oppenheimer truncation reduces it to (4.21), giving the familiar single-PES picture (4.22).

Appendix

Appendix A: Mass-Polarization Term in Classical Molecular Hamiltonian

We start from the laboratory-frame kinetic energy

$$T = \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j^2 + \frac{1}{2} \sum_i m_e \dot{\mathbf{r}}_i^2, \quad (1)$$

where M_j is the mass of the j -th nucleus, $\mathbf{R}_j$ is its Cartesian coordinate, m_e is the electron mass, and $\mathbf{r}_i$ is the coordinate of the i -th electron.

4.7.6 Nuclear Center-of-Mass Coordinate

Define the nuclear total mass

$$M_N = \sum_j M_j. \quad (2)$$

The nuclear center-of-mass coordinate is

$$\mathbf{R}_{\text{cm}} = \mathbf{R}_N^{\text{CoM}} = \frac{1}{M_N} \sum_j M_j \mathbf{R}_j. \quad (3)$$

We introduce coordinates shifted by the nuclear center of mass:

$$\mathbf{R}'_j = \mathbf{R}_j - \mathbf{R}_{\text{cm}}, \quad (4)$$

$$\mathbf{r}'_i = \mathbf{r}_i - \mathbf{R}_{\text{cm}}. \quad (5)$$

Therefore,

$$\mathbf{R}_j = \mathbf{R}_{\text{cm}} + \mathbf{R}'_j, \quad \mathbf{r}_i = \mathbf{R}_{\text{cm}} + \mathbf{r}'_i. \quad (6)$$

The shifted nuclear coordinates satisfy the constraint

$$\sum_j M_j \mathbf{R}'_j = 0, \quad (7)$$

and therefore

$$\sum_j M_j \dot{\mathbf{R}}'_j = 0. \quad (8)$$

This is important: the $\mathbf{R}'_j$ are not all independent.

4.7.7 Kinetic Energy in Velocity Form

Substitute

$$\dot{\mathbf{R}}_j = \dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j, \quad \dot{\mathbf{r}}_i = \dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \quad (9)$$

into the kinetic energy. For the nuclear part,

$$T_N = \frac{1}{2} \sum_j M_j \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{R}}'_j \right)^2 \quad (10)$$

$$= \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2 + \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_j M_j \dot{\mathbf{R}}'_j + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2. \quad (11)$$

Using

$$\sum_j M_j \dot{\mathbf{R}}'_j = 0, \quad (12)$$

we get

$$T_N = \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2 + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2. \quad (13)$$

For the electronic part,

$$T_e = \frac{1}{2} \sum_i m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \right)^2 \quad (14)$$

$$= \frac{1}{2} N_{\text{elec}} m_e \dot{\mathbf{R}}_{\text{cm}}^2 + m_e \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2} \sum_i m_e \dot{\mathbf{r}}_i'^2, \quad (15)$$

where N_{elec} is the number of electrons. Hence the total kinetic energy is

$$T = \frac{1}{2} (M_N + N_{\text{elec}} m_e) \dot{\mathbf{R}}_{\text{cm}}^2 + m_e \dot{\mathbf{R}}_{\text{cm}} \cdot \sum_i \dot{\mathbf{r}}'_i + \frac{1}{2} \sum_j M_j \dot{\mathbf{R}}_j'^2 + \frac{1}{2} \sum_i m_e \dot{\mathbf{r}}_i'^2. \quad (16)$$

4.7.8 Canonical Momenta

The momentum conjugate to the nuclear center-of-mass coordinate $\mathbf{R}_{\text{cm}}$ is

$$\mathbf{P}_{\text{cm}} = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{R}}_{\text{cm}}} \quad (17)$$

$$= (M_N + N_{\text{elec}} m_e) \dot{\mathbf{R}}_{\text{cm}} + m_e \sum_i \dot{\mathbf{r}}'_i. \quad (18)$$

Since

$$\mathbf{p}'_i = \frac{\partial \mathcal{L}}{\partial \dot{\mathbf{r}}'_i} = m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \right), \quad (19)$$

we can write

$$\sum_i \mathbf{p}'_i = N_{\text{elec}} m_e \dot{\mathbf{R}}_{\text{cm}} + m_e \sum_i \dot{\mathbf{r}}'_i. \quad (20)$$

Therefore,

$$\boxed{\mathbf{P}_{\text{cm}} = M_N \dot{\mathbf{R}}_{\text{cm}} + \sum_i \mathbf{p}'_i} \quad (21)$$

or equivalently,

$$\boxed{M_N \dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i} \quad (22)$$

and hence

$$\dot{\mathbf{R}}_{\text{cm}} = \frac{1}{M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i \right). \quad (23)$$

The electron momentum conjugate to $\mathbf{r}'_i$ is

$$\mathbf{p}'_i = m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}'_i \right). \quad (24)$$

Thus $\mathbf{p}'_i$ is the lab-frame electron momentum, not simply $m_e \dot{\mathbf{r}}'_i$. Equivalently,

$$m_e \dot{\mathbf{r}}'_i = \mathbf{p}'_i - m_e \dot{\mathbf{R}}_{\text{cm}}. \quad (25)$$

The nuclear shifted momenta are

$$\mathbf{P}'_j = M_j \dot{\mathbf{R}}'_j. \quad (26)$$

Because the shifted nuclear coordinates satisfy

$$\sum_j M_j \mathbf{R}'_j = 0, \quad (27)$$

their conjugate momenta satisfy

$$\sum_j \mathbf{P}'_j = 0. \quad (28)$$

In terms of the laboratory-frame nuclear momenta

$$\mathbf{P}_j = M_j \dot{\mathbf{R}}_j, \quad (29)$$

one has

$$\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N} \mathbf{P}_N, \quad (30)$$

where

$$\mathbf{P}_N = \sum_j \mathbf{P}_j = M_N \dot{\mathbf{R}}_{\text{cm}}. \quad (31)$$

Using

$$\mathbf{P}_N = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i, \quad (32)$$

we may also write

$$\mathbf{P}'_j = \mathbf{P}_j - \frac{M_j}{M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}'_i \right). \quad (33)$$

4.7.9 Kinetic Energy in Momentum Form

The original kinetic energy can be written as

$$T = T_{\text{nuc,int}} + T_{\text{nuc,CoM}} + T_e. \quad (34)$$

The nuclear internal kinetic energy is

$$T_{\text{nuc,int}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j}. \quad (35)$$

The electron kinetic energy is

$$T_e = \sum_i \frac{\mathbf{p}_i'^2}{2m_e}. \quad (36)$$

The nuclear center-of-mass kinetic energy is

$$T_{\text{nuc,CoM}} = \frac{1}{2} M_N \dot{\mathbf{R}}_{\text{cm}}^2. \quad (37)$$

Using

$$M_N \dot{\mathbf{R}}_{\text{cm}} = \mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i', \quad (38)$$

we get

$$T_{\text{nuc,CoM}} = \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2. \quad (39)$$

Therefore the kinetic energy in terms of the new momenta is

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2. \quad (40)$$

The accompanying constraints are

$$\sum_j M_j \mathbf{R}_j' = 0, \quad \sum_j \mathbf{P}_j' = 0. \quad (41)$$

4.7.10 Expanded Form

Expanding the center-of-mass recoil term gives

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}_i' + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (42)$$

Thus,

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{\mathbf{P}_{\text{cm}}^2}{2M_N} - \frac{\mathbf{P}_{\text{cm}}}{M_N} \cdot \sum_i \mathbf{p}_i' + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (43)$$

4.7.11 Hamiltonian

If the potential energy is translationally invariant, then

$$V = V(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\}). \quad (44)$$

The Hamiltonian is therefore

$$H = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2 + V(\{\mathbf{R}_j'\}, \{\mathbf{r}_i'\}). \quad (45)$$

4.7.12 Molecular Rest Frame

If the whole molecule is taken to be in its rest frame, then

$$\mathbf{P}_{\text{cm}} = 0. \quad (46)$$

The kinetic energy becomes

$$T_{\text{rest}} = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2. \quad (47)$$

The last term,

$$\frac{1}{2M_N} \left(\sum_i \mathbf{p}_i' \right)^2, \quad (48)$$

is the nuclear center-of-mass recoil term, also called the mass-polarization term.

4.7.13 Important Interpretation

The coordinate transformation used here shifts both nuclei and electrons by the nuclear center of mass, not by the total molecular center of mass. Therefore,

$$\mathbf{p}_i' = m_e \left(\dot{\mathbf{R}}_{\text{cm}} + \dot{\mathbf{r}}_i' \right) \quad (49)$$

is the canonical momentum conjugate to $\mathbf{r}_i'$. It is equal to the electron lab-frame mechanical momentum. However,

$$m_e \dot{\mathbf{r}}_i' \quad (50)$$

is only the electron momentum relative to the moving nuclear center of mass. It is not the canonical momentum conjugate to $\mathbf{r}_i'$ unless the nuclear center-of-mass motion has been fixed or eliminated. The final kinetic energy in nuclear-CoM coordinates is therefore

$$T = \sum_j \frac{\mathbf{P}_j'^2}{2M_j} + \sum_i \frac{\mathbf{p}_i'^2}{2m_e} + \frac{1}{2M_N} \left(\mathbf{P}_{\text{cm}} - \sum_i \mathbf{p}_i' \right)^2, \quad (51)$$

with

$$\sum_j M_j \mathbf{R}_j' = 0, \quad \sum_j \mathbf{P}_j' = 0. \quad (52)$$

Appendix B: Full Derivation of the Watson Pseudopotential

Here we supply the reduction quoted between (4.10c) and (4.11): how curvilinear quantization of the Watson Hamiltonian (3.22), after the rescaling (4.5), produces the compact

pseudopotential $\hat{U} = -\frac{\hbar^2}{8} \sum_{\alpha} \mu_{\alpha\alpha}$.

B.1 The general curvilinear extrapotential

Let q^A be arbitrary coordinates with metric g_{AB} , determinant $g = \det g_{AB}$, and classical kinetic energy $2T = g^{AB} p_A p_B$. The unique kinetic operator Hermitian on $L^2(\sqrt{g} d^n q)$ is the Laplace–Beltrami operator

$$\hat{T}_{\text{LB}} = -\frac{\hbar^2}{2} g^{-1/2} \partial_A (g^{1/2} g^{AB} \partial_B). \quad (\text{B.1})$$

To pass to a flat measure $\prod_A dq^A$ we rescale $\psi_W = g^{1/4} \psi$, i.e. $\hat{T}_W = g^{1/4} \hat{T}_{\text{LB}} g^{-1/4}$. With $\rho \equiv g^{1/2}$ and $f_B \equiv \partial_B \psi_W - \frac{1}{2} (\partial_B \ln \rho) \psi_W$,

$$\hat{T}_W \psi_W = -\frac{\hbar^2}{2} \left[\frac{1}{2} (\partial_A \ln \rho) g^{AB} f_B + (\partial_A g^{AB}) f_B + g^{AB} \partial_A f_B \right]. \quad (\text{B.2})$$

The terms linear in $\partial \psi_W$ are $\frac{1}{2} g^{AB} (\partial_A \ln \rho) \partial_B \psi_W$ (first bracket) and $-\frac{1}{2} g^{AB} (\partial_B \ln \rho) \partial_A \psi_W$ (third bracket); by symmetry of g^{AB} they cancel, while the second-derivative part re-assembles into the flat symmetric operator $\frac{1}{2} \hat{p}_A g^{AB} \hat{p}_B$ ($\hat{p}_A = -i\hbar \partial_A$). The surviving ψ_W -proportional pieces define the *extrapotential*; using $\ln \rho = \frac{1}{2} \ln g$,

$$\boxed{\hat{T}_W = \frac{1}{2} \hat{p}_A g^{AB} \hat{p}_B + U, \quad U = \frac{\hbar^2}{8} \left[\frac{1}{4} g^{AB} \partial_A \ln g \partial_B \ln g + \partial_A (g^{AB} \partial_B \ln g) \right].} \quad (\text{B.3})$$

Equation (B.3) is exact and coordinate-general; the pseudopotential is the entire c -number U , and the prefactor $\frac{\hbar^2}{8}$ traces to the two half-powers $g^{\pm 1/4}$ of the rescaling.

B.2 The molecular metric

For (3.22) the coordinates are $q^A = (\theta, \phi, \chi, Q_b)$. By (4.4), $\sqrt{|g|} = \sin \theta (\det \mathbf{I}')^{1/2}$, so

$$\ln g = 2 \ln \sin \theta + \ln \gamma, \quad \gamma \equiv \det \mathbf{I}'(Q) = 1 / \det \boldsymbol{\mu}, \quad (\text{B.4})$$

and all Q -dependence of the measure sits in γ . Reading the contravariant metric off (3.22) in the body-frame momentum basis $(\hat{J}_\alpha, \hat{P}_b)$ [with $\hat{\pi}_\alpha = \sum_b \sigma_{b\alpha} \hat{P}_b$, $\sigma_{b\alpha} = (\boldsymbol{\sigma}_b)_\alpha$],

$$\mathbf{g}^{-1} = \begin{pmatrix} \boldsymbol{\mu} & -\boldsymbol{\mu} \boldsymbol{\sigma}^\top \\ -\boldsymbol{\sigma} \boldsymbol{\mu} & \mathbb{1} + \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\sigma}^\top \end{pmatrix}, \quad \det \mathbf{g}^{-1} = \det \boldsymbol{\mu} = 1/\gamma, \quad (\text{B.5})$$

the determinant following from the Schur reduction $\det(\mathbb{1} + \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\sigma}^\top - \boldsymbol{\sigma} \boldsymbol{\mu} \boldsymbol{\mu}^{-1} \boldsymbol{\mu} \boldsymbol{\sigma}^\top) = 1$, consistent with (B.4).

B.3 Reduction to the trace

In the rovib convention the body-frame angular momenta $\hat{J}_\alpha$ already carry $\sin \theta$ (they are Hermitian on $\sin \theta d\Omega$, with the anomalous algebra (4.7)), so only the residual weight $\gamma^{1/2}(Q)$ is removed by the rescaling (4.5). Three groups of terms arise.

(i) *Rotational.* In $\frac{1}{2} \hat{J}_\alpha \mu_{\alpha\beta} \hat{J}_\beta$ the factor $\mu_{\alpha\beta}(Q)$ commutes with $\hat{J}$ and is symmetric; writing $\frac{1}{2} \mu_{\alpha\beta} \hat{J}_\alpha \hat{J}_\beta = \frac{1}{4} \mu_{\alpha\beta} \{\hat{J}_\alpha, \hat{J}_\beta\} + \frac{1}{4} \mu_{\alpha\beta} [\hat{J}_\alpha, \hat{J}_\beta]$ and using the anomalous $[\hat{J}_\alpha, \hat{J}_\beta] =$

$-i\hbar\epsilon_{\alpha\beta\gamma}\hat{J}_\gamma$ with $\mu_{\alpha\beta}\epsilon_{\alpha\beta\gamma} = 0$, the commutator drops: the asymmetric-top operator is Hermitian as written and yields *no* c -number.

(ii) *Vibrational*. The diagonal $\hat{P}_b^2$ part, with weight $\gamma^{1/2}$, gives $\frac{1}{2}\sum_b \hat{P}_b^2$ plus the Pauli residue [exactly (4.10c)],

$$U_{\text{vib}} = \frac{\hbar^2}{2} \gamma^{-1/4} \left(\sum_b \partial_b^2 \right) \gamma^{1/4} = \frac{\hbar^2}{8} \sum_b \left[\partial_b^2 \ln \gamma + \frac{1}{4} (\partial_b \ln \gamma)^2 \right]. \quad (\text{B.6})$$

(iii) *Coriolis*. The off-diagonal $-\hat{J}_\alpha(\boldsymbol{\mu}\boldsymbol{\sigma}^\text{T})_{\alpha b}\hat{P}_b$ block, together with the γ -dressing of $\hat{P}_b$ and the commutators $[\hat{P}_b, \mu_{\alpha\beta}]$ and $[\hat{P}_b, \sigma_{a\alpha}]$, supplies the remaining Q -derivative c -numbers.

The collapse rests on two exact identities. Jacobi's formula for $\gamma = \det \mathbf{I}'$,

$$\partial_b \ln \gamma = \text{tr}(\boldsymbol{\mu} \partial_b \mathbf{I}'), \quad \partial_b \boldsymbol{\mu} = -\boldsymbol{\mu} (\partial_b \mathbf{I}') \boldsymbol{\mu}, \quad (\text{B.7})$$

rewrites every $\partial_b \ln \gamma$ and $\partial_b \boldsymbol{\mu}$ appearing in (B.6) and (iii) through $\partial_b \mathbf{I}'$; and because $\mathbf{I}' = \mathbf{I}_N - \boldsymbol{\sigma}\boldsymbol{\sigma}^\text{T}$ with $\boldsymbol{\sigma}_b = \sum_a \boldsymbol{\zeta}_{ab} Q_a$ of (3.17a) *linear* in Q , the derivatives $\partial_b \mathbf{I}'$ are built from the *constant* Coriolis matrices $\boldsymbol{\zeta}_{ab}$, whose contractions satisfy the orthogonality sum rules of the Eckart mode vectors ($\sum_j \mathbf{l}_{ja} \cdot \mathbf{l}_{jb} = \delta_{ab}$; the $\boldsymbol{\zeta}^{(\alpha)}$ are the antisymmetric $SO(3)$ generators on the mode index). Substituting (B.7) into groups (ii) and (iii), the gradient terms $\propto \partial_b \text{tr} \boldsymbol{\mu}$ produced by the vibrational residue are cancelled identically by those of the Coriolis block, and the surviving Q -local remainder is the single trace

$$\boxed{\hat{U} = U_{\text{rot}} + U_{\text{vib}} + U_{\text{Cor}} = -\frac{\hbar^2}{8} \sum_{\alpha=1}^3 \mu_{\alpha\alpha}(Q)} \quad (\text{B.8})$$

of (4.11). This reproduces Watson's reduction [J. K. G. Watson, *Mol. Phys.* **15**, 479 (1968)]. The cancellation is special to rectilinear normal coordinates: for general curvilinear Q_b the residue (B.6) does *not* collapse to the trace, leaving the additional vibrational extrapotential terms anticipated in Section 4.1. The electronic sector contributes nothing to $\hat{U}$, since $\boldsymbol{\mu}$, γ , and $\boldsymbol{\sigma}$ depend on Q alone.

References